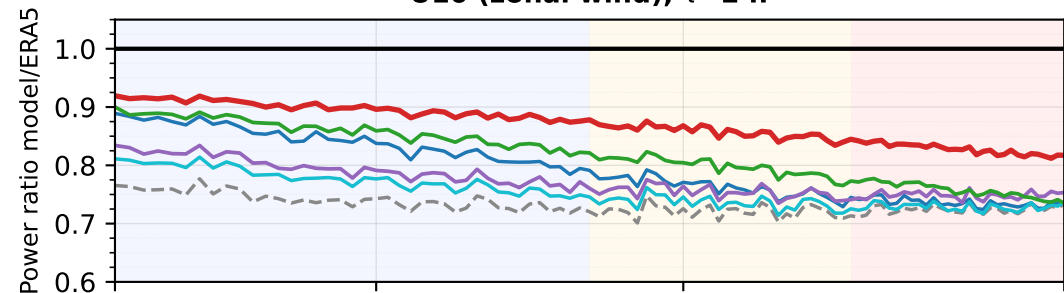
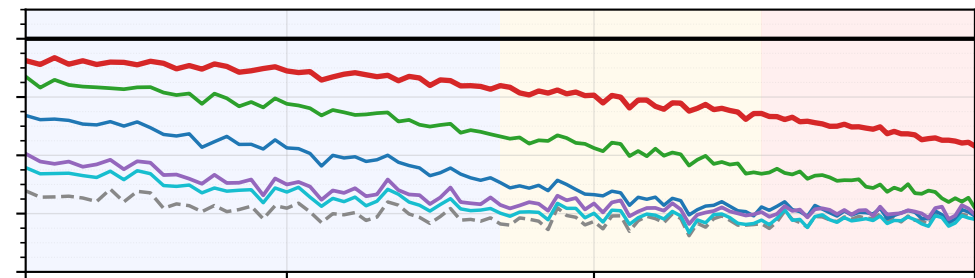


Spherical-harmonic power ratio model/ERA5 ($\lambda \in [2.0^\circ, 4.5^\circ]$, $\ell \in [80, 180]$)

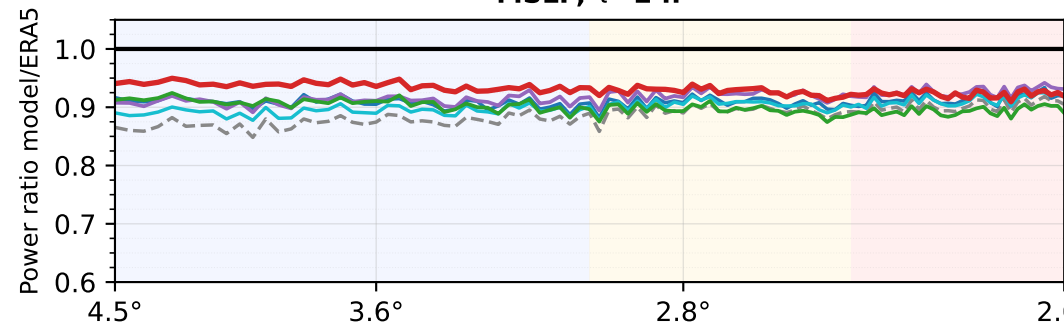
U10 (zonal wind), $\tau=2$ h



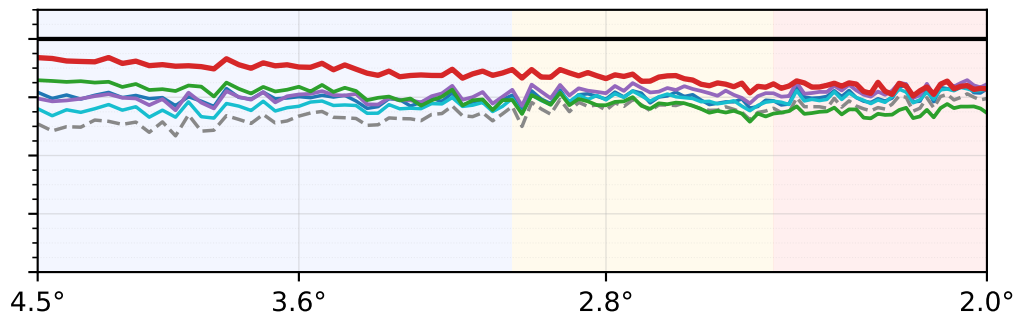
U10 (zonal wind), $\tau=3$ h



MSLP, $\tau=2$ h



MSLP, $\tau=3$ h



Angular wavelength $\lambda = 360^\circ / \ell$ (synoptic $\rightarrow$ meso)

--- Linear Interp.

— ModAFNO

— S-DYff

— ATM-VFI

— WeatherDCAE-NoSkip (37M)

— FuXi